3 REAL-TIME CARDIAC MONITOR The Size Of A Band-Aid

Medical electronics need to be designed in a manner that these not only provide accurate health data but are also easy to use. This enables quick and efficient diagnosis of ailments. Keeping this in mind, Taiwan-based Singular Wings Medical has developed a tracking device for cardiovascular diseases the size of a bandaid, called CardiNova Heart Disease Recorder.

The wireless non-intrusive device can be connected to the chest using the patch-like fabric on which the analyser is mounted. There is a button at the front of the device, which, when pressed, starts device operation. Two electrodes (visible on back) pick up cardiovascular readings as electrical signals. A microprocessor unit processes signal data.

Results can be seen in real time over a proprietary mobile application. The proprietary algorithm helps eliminate the noise created by motion, contact deviation and the like, and pick up precise signal readings. Data provides quick reference information for health, including heart beat rate, fitness level, user mood and behaviour, cardiac rhythm information and more.

While wearing CardiNova, the user can turn on monitoring at the slightest feeling of discomfort, so that any impending problem can be identified, and immediate action taken.

"We minimised the form factor and made setting up of the device easy. A family member or doctor can monitor data and identify a problem as soon as it occurs. The algorithm can summarise the report by learning from huge amounts of data accumulated from long-term measurement. It eases the process of detecting the main symptom directly. It also raises an alarm and sends notifications to stored contacts in case of an emergency, so that immediate action can be taken," explains a spokesperson from Singular Wings Medical.

The spokesperson adds, "The system has been tested to ensure that sensors pick up signals accurately. A commercial chip is used for processing the data. Bluetooth module enables connectivity with a smartphone."

Data can also be stored on a commercial cloud platform.

The company has also brought forward a full line of smartwear suits, with CardiNova monitor embedded in the chest area. Fabric of the suit is elastic, and it soaks and dries moisture, and keeps the user comfortable. Area of fabric around the monitoring device can sense and transmit cardiovascular signals.

CardiNova can be highly useful for those with pre-existing cardiovascular diseases or those with a surgical history. Hospitals can monitor patient health in real time with the device. It can also be used for other purposes. For example, a sportsperson can monitor his/her fitness condition, or an end user can stay aware about his/her general physical health. Singular Wings Medical is selling CardiNova through their distributor network globally.

4 SELF-DRIVING CAR To Give You A Lift

Autonomous cars are finally on their way. Experts predict that within a span of five years, we will see self-driving vehicles become mainstream on the roads. Automotive players around the world are leveraging on this opportunity with their autonomous driving solutions. One such innovative solution is DragonFly autonomous vehicle, created by PerceptIn.

Dr Shaoshan Liu, founder and chairman, PerceptIn, created DragonFly to make low-cost autonomous vehicles available for use. His vision was to create self-driven vehicle that was affordable and could aid the elderly, physically-challenged or those who could not drive. Utilising his previous experience with automotive artificial intelligence (AI) programs and sen-